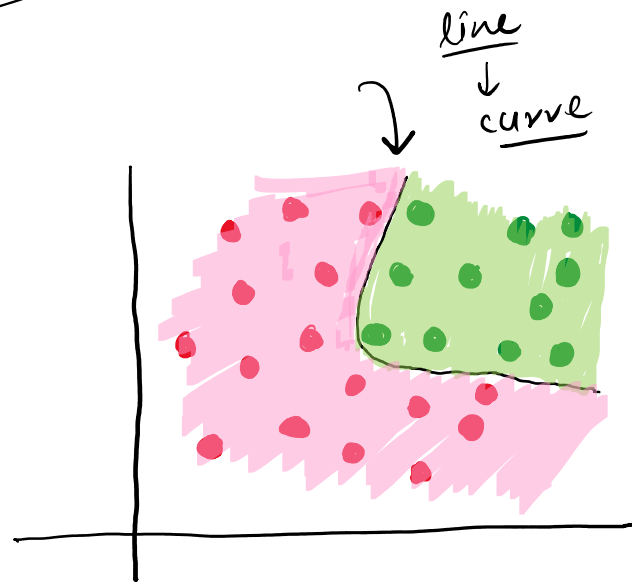
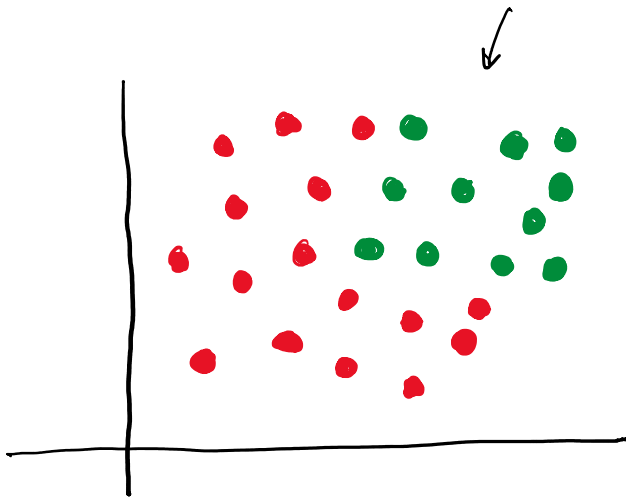


The Problem

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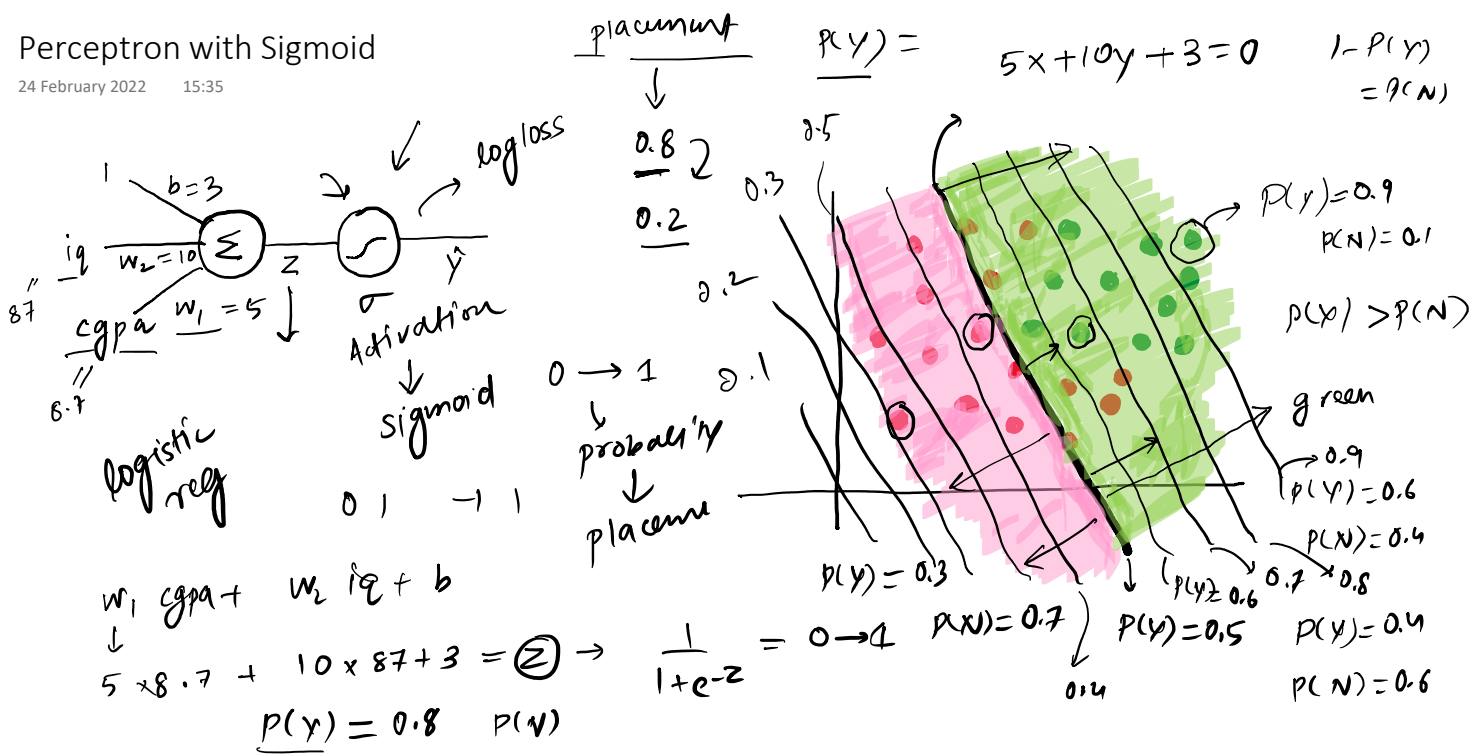
perceptron



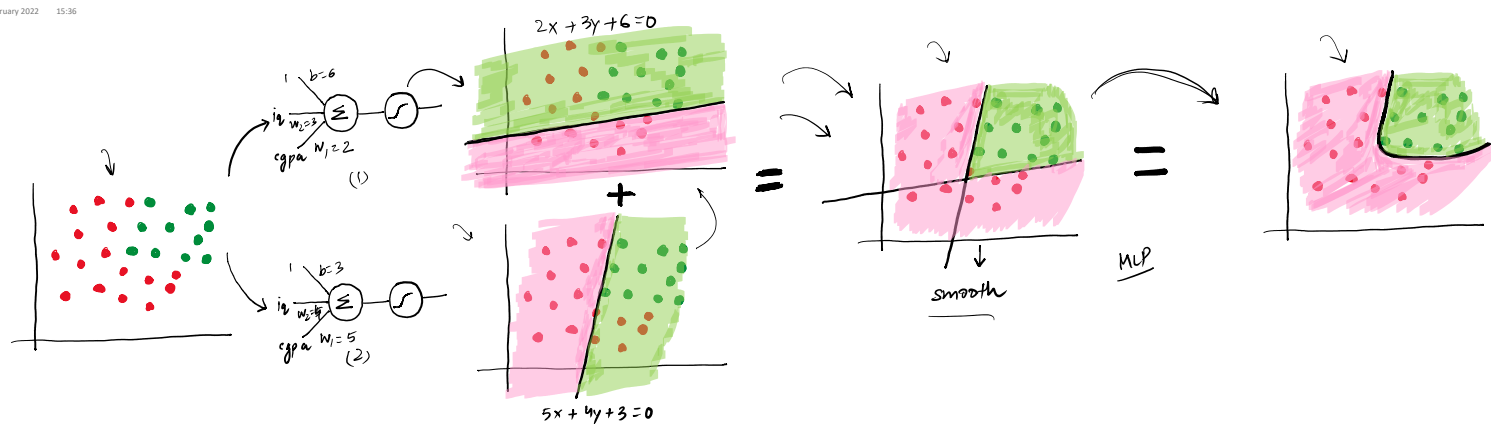
- > We already know the main problem with the perceptron trick
- > It can't classify non linear data
- > So, we need something that can draw curve on the data based on the non linearity

Perceptron with Sigmoid

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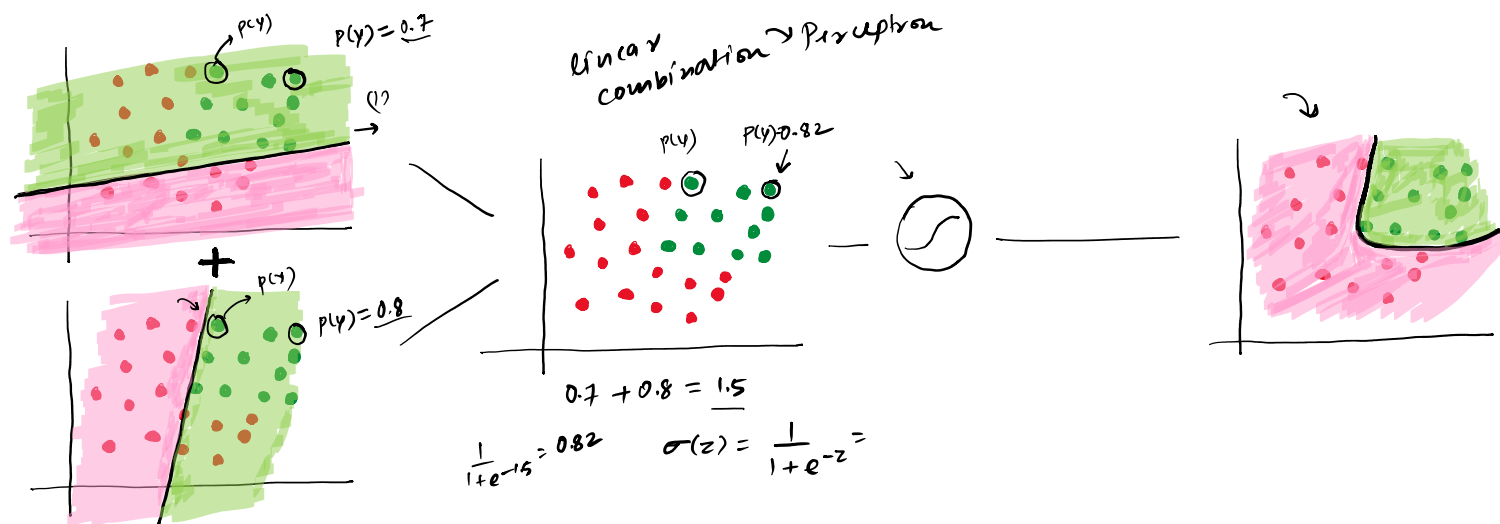


- > We also know how sigmoid function works on perceptron trick
- > We place the data on the equation of the line and we pass it to the sigmoid function and it will tell us the probability of being 1



-> Here is a idea, how i can be done to make a curve using two perceptron trick seperately and fuse them together and after smoothing we will get the perfect curve.

-> From here we can get a idea, how the ann works.



-> Let's see, how it is done by mathematics

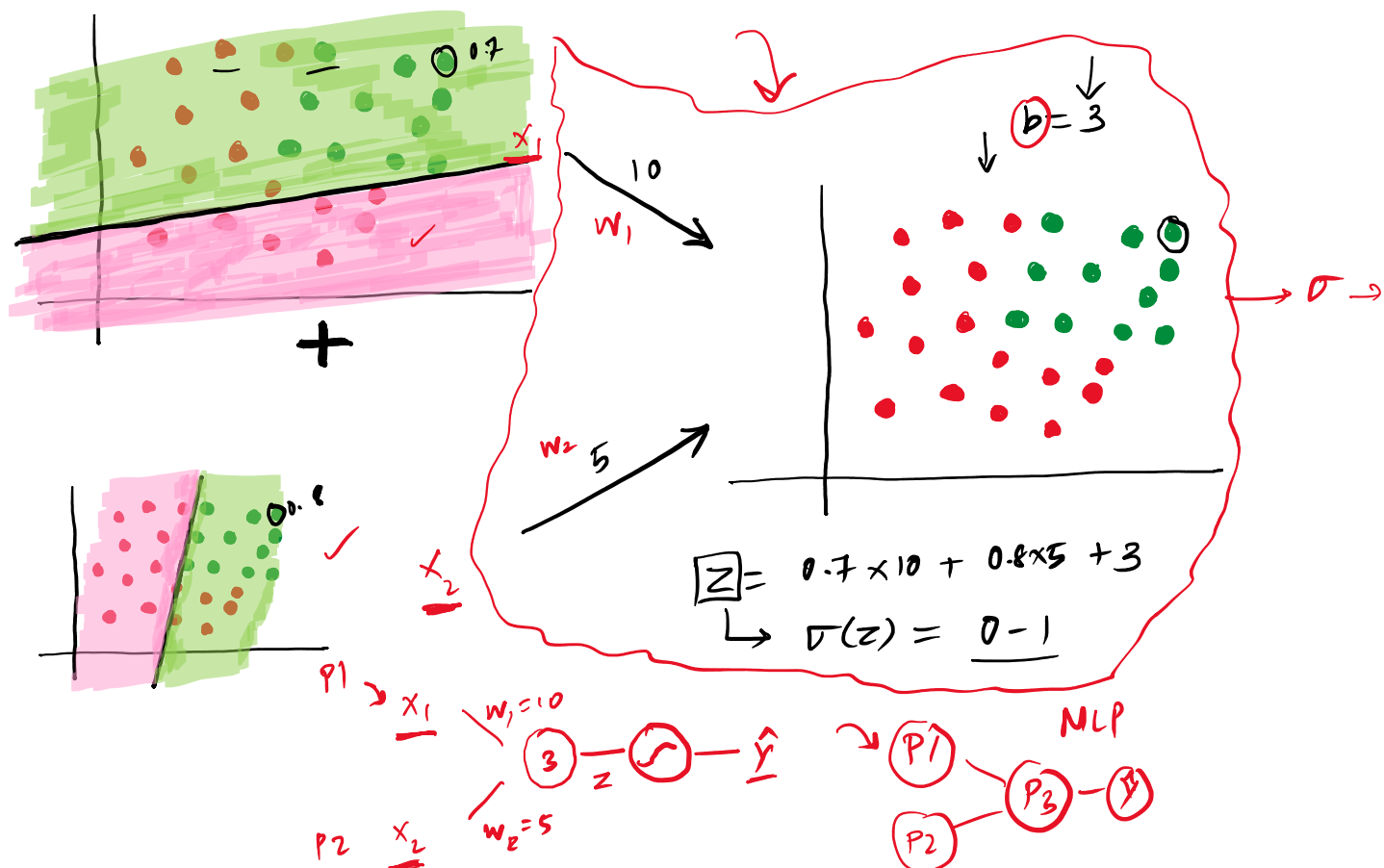
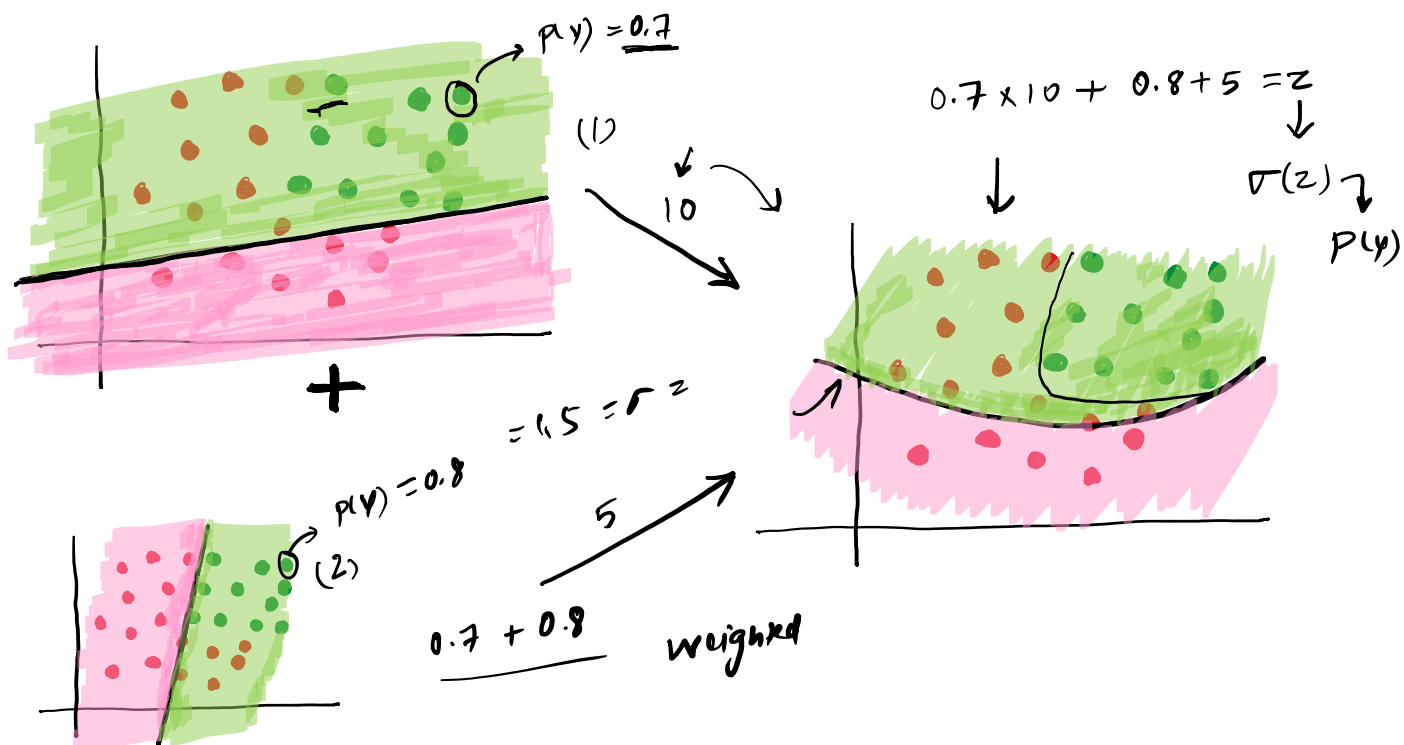
-> For each point/ student we will get 2 output from two perceptron.

-> So, we will combine those two output together and pass to the sigmoid again, so that we can get a prediction with the range of 0-1.

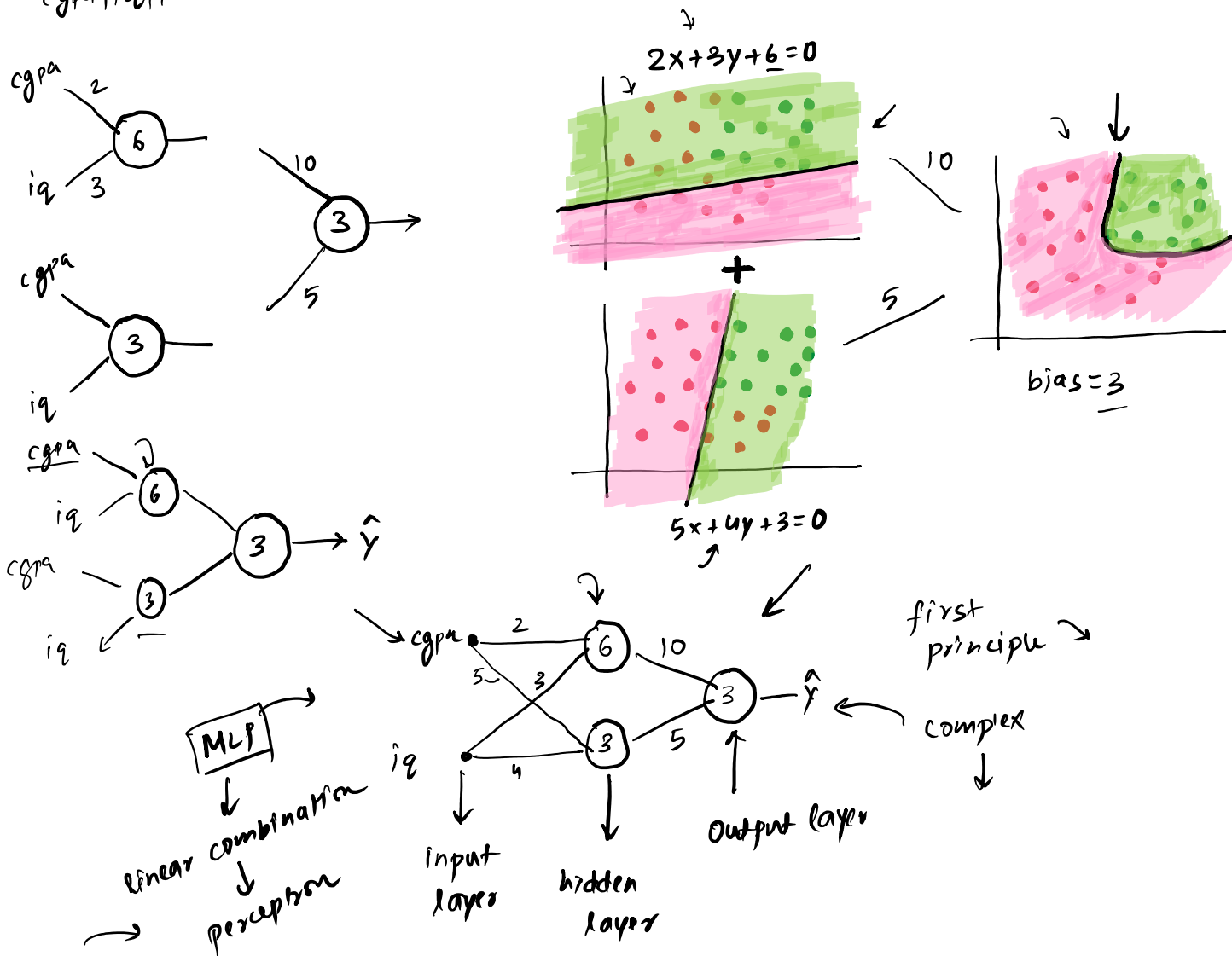
-> We will perform this on every point and we will get the curve.

Adding Weights

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cgpa | iq | place



-> 1st: Let's say, we want a graph like the picture. So, what we can do? to make the graph most likely top one than the bottom one.

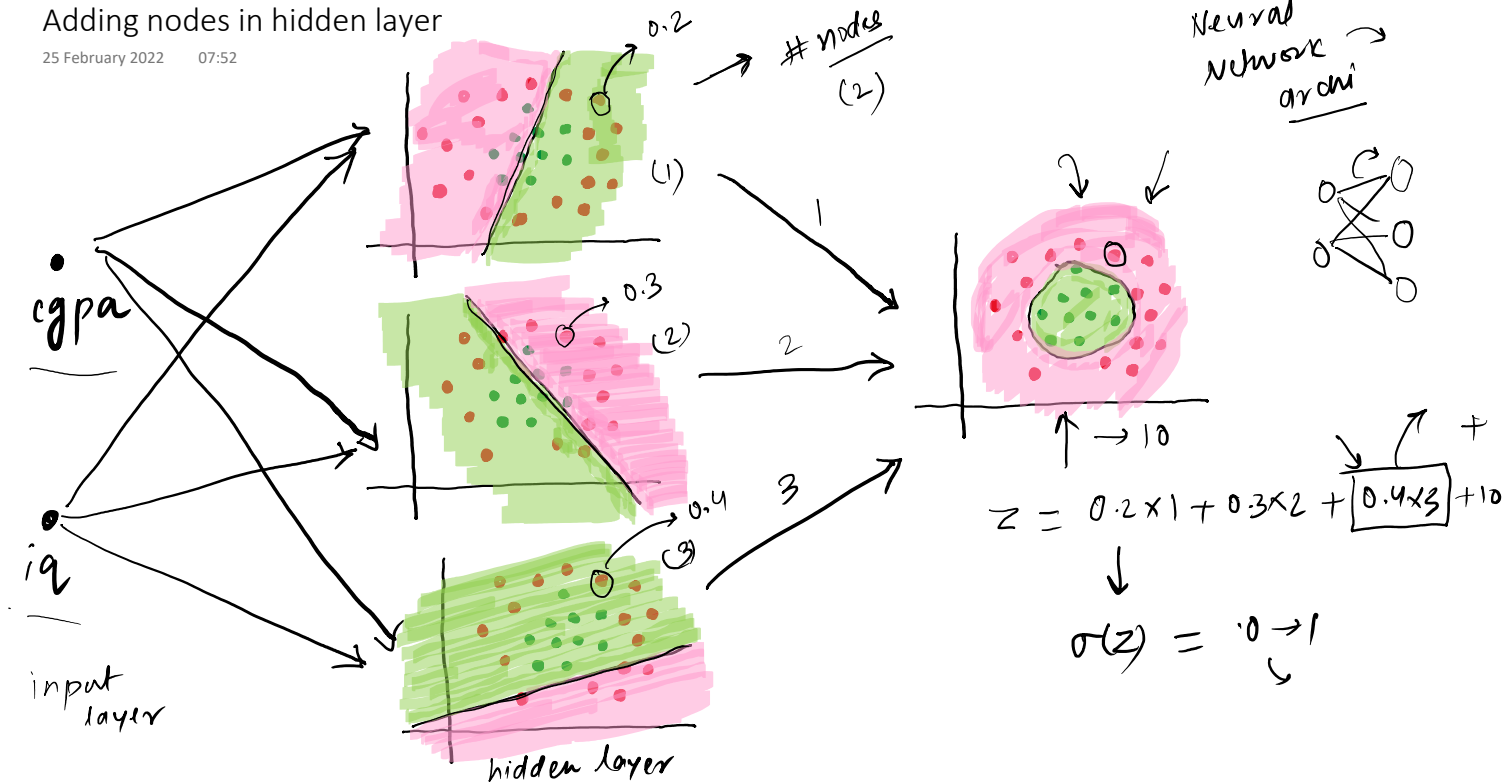
-> we can add weights to both of them and assign high weights to the top one than bottom so that the graph becomes most likely to the top one.

-> 2nd: Here, we are actually doing the same perceptron trick on second part same as 1st two perceptrons. That's why it is called MLP. In this example the layer is 2.

-> 3rd: We can see the full intuition here, how it becomes MLP.

Adding nodes in hidden layer

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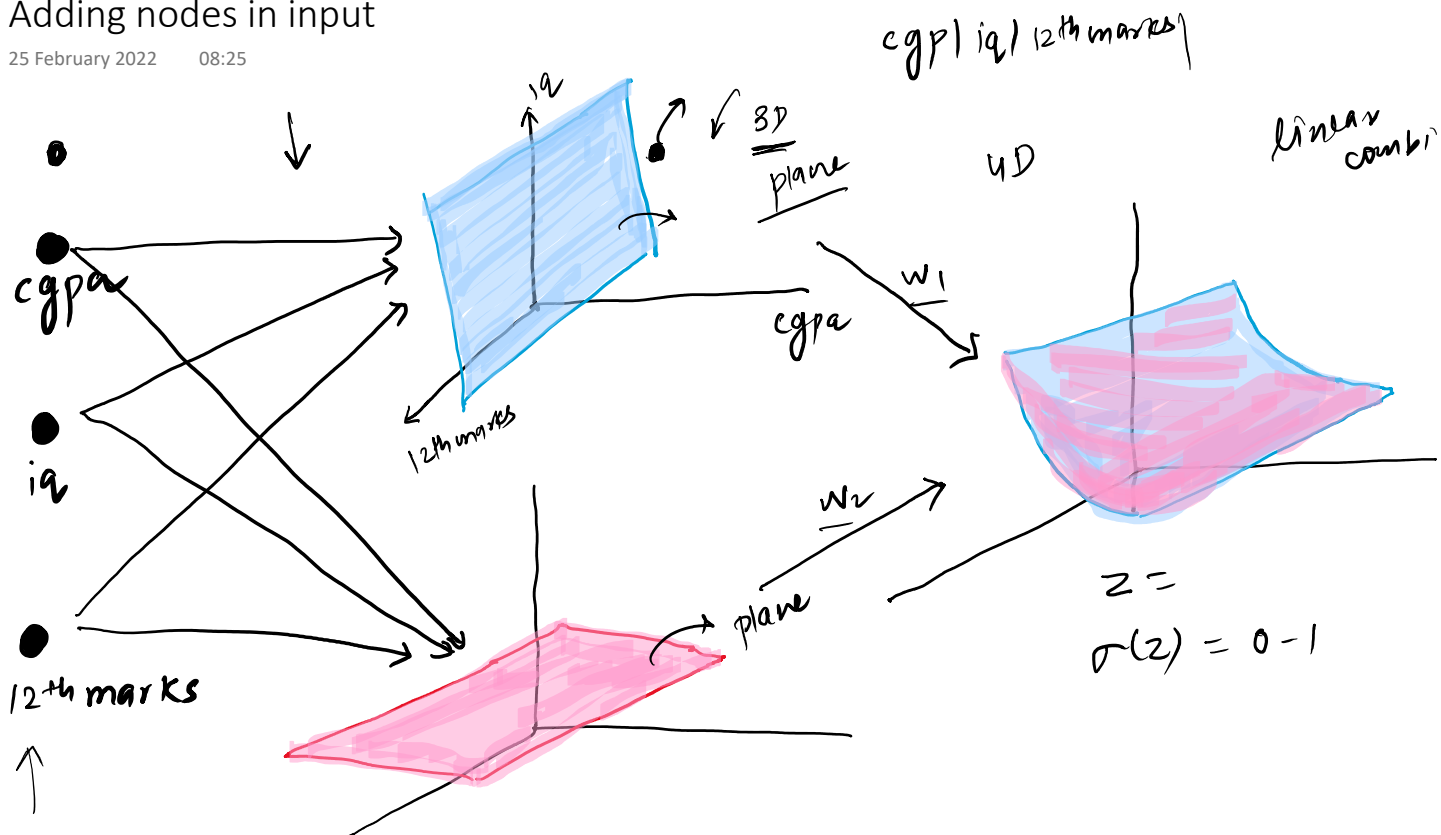


-> By changing the structure of MLP we can meet our certain requirements.

-> In this example, we add another node on the hidden layer so that it can capture more feature and can give us the final structure that we want to achieve.

Adding nodes in input

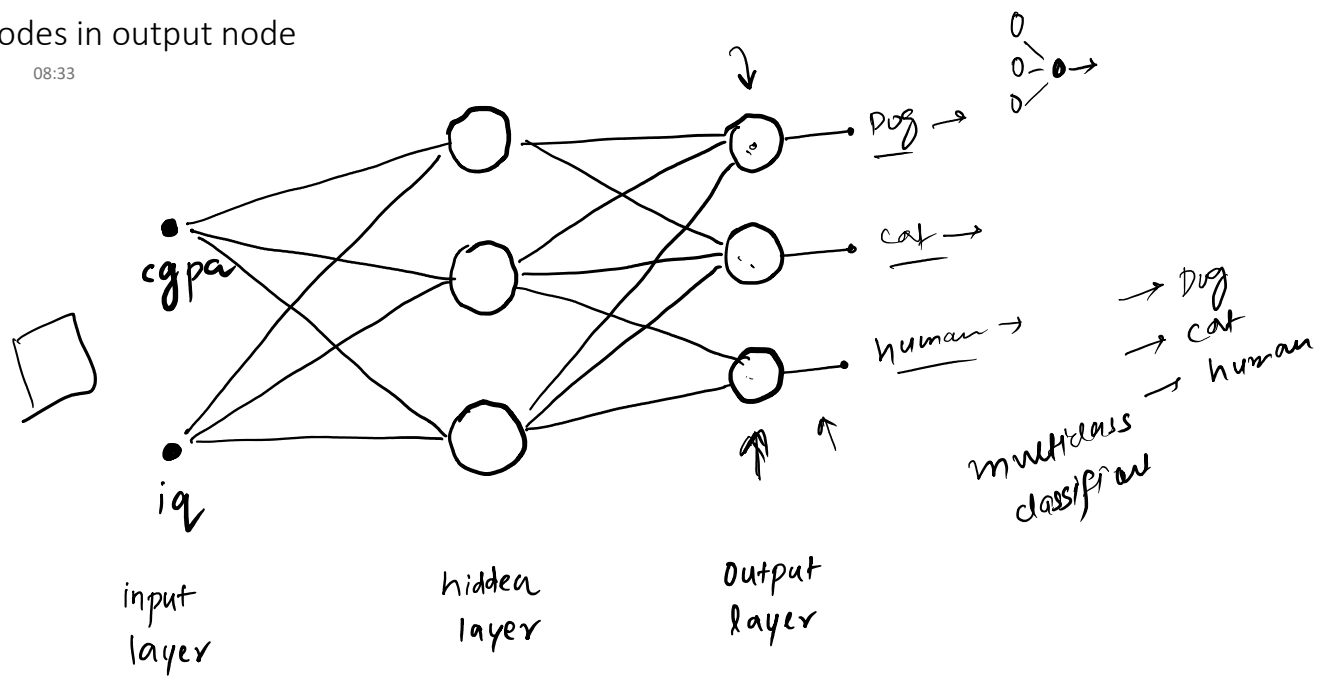
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- > If we have multiple columns on the data. We can also add them to the structure, if the column carry good feature.
- > By adding more node to the input layer we are actually adding more dimension.
- > Here the 2D strcture becomes 3D after adding a extra column. and now the dicision bounday is a plain rather than a line. After a hidden layer we will get a curve plane that we wanted to achieve.
- > If we add another input the decision bounday will become a box, or hyper plane.

Adding nodes in output node

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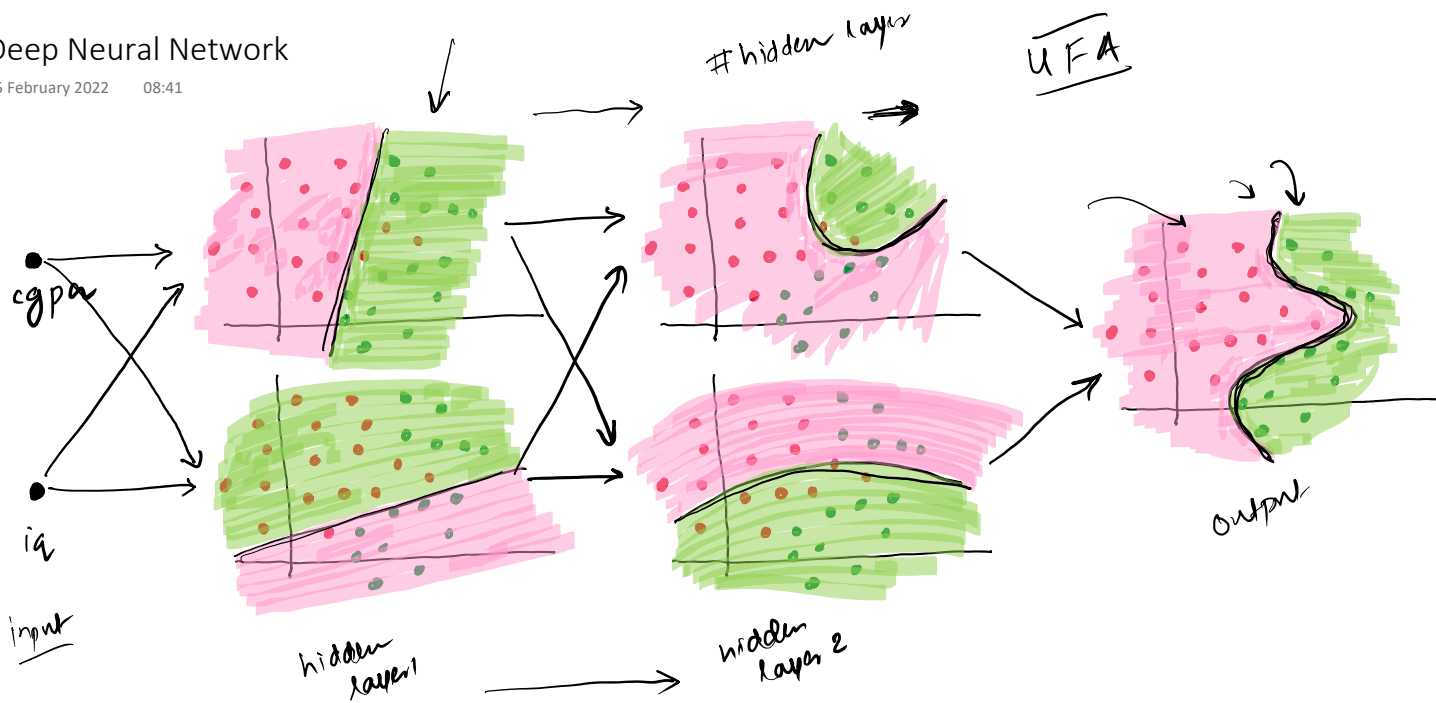


-> We can add node on output.

-> Generally we are doing this when we need multiclass classification.

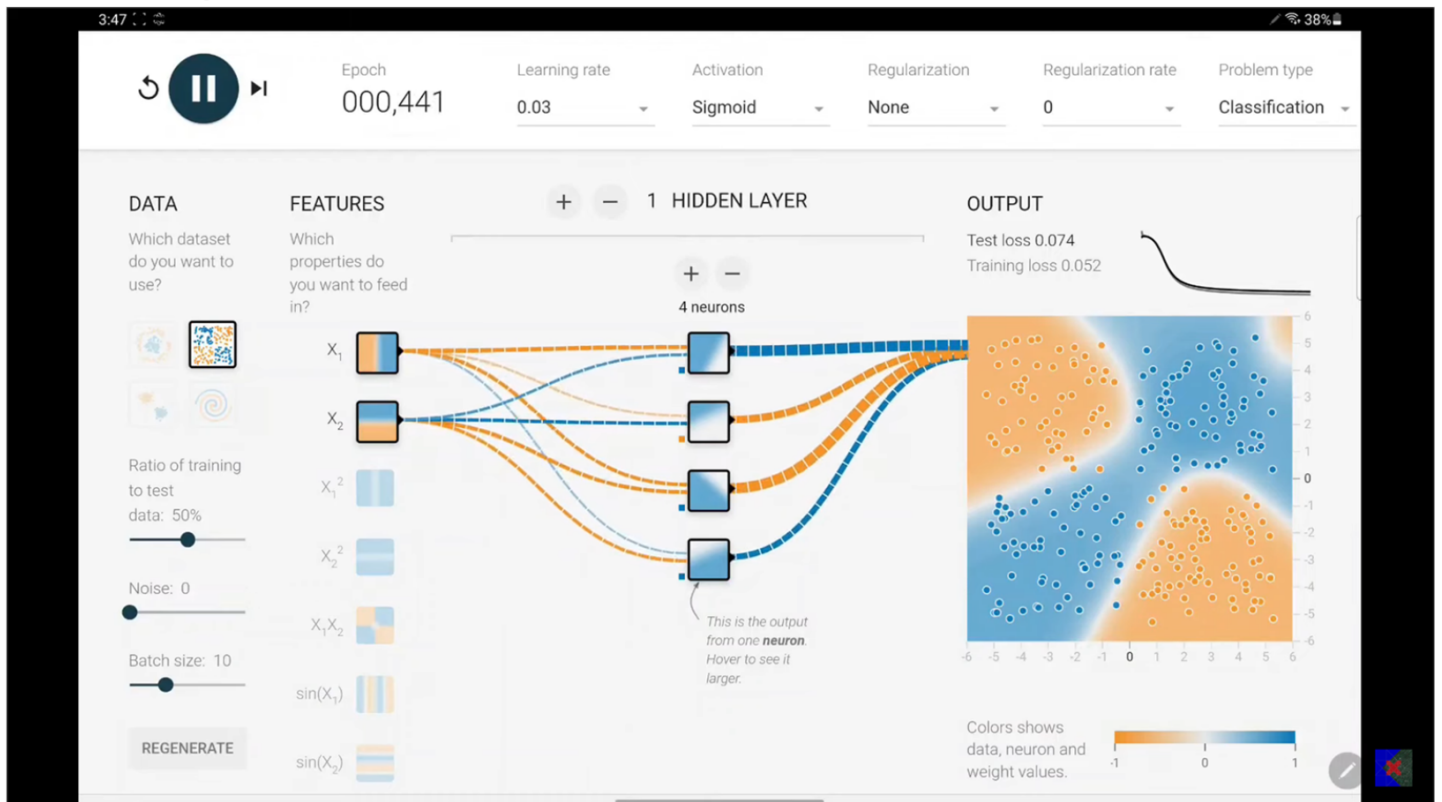
Deep Neural Network

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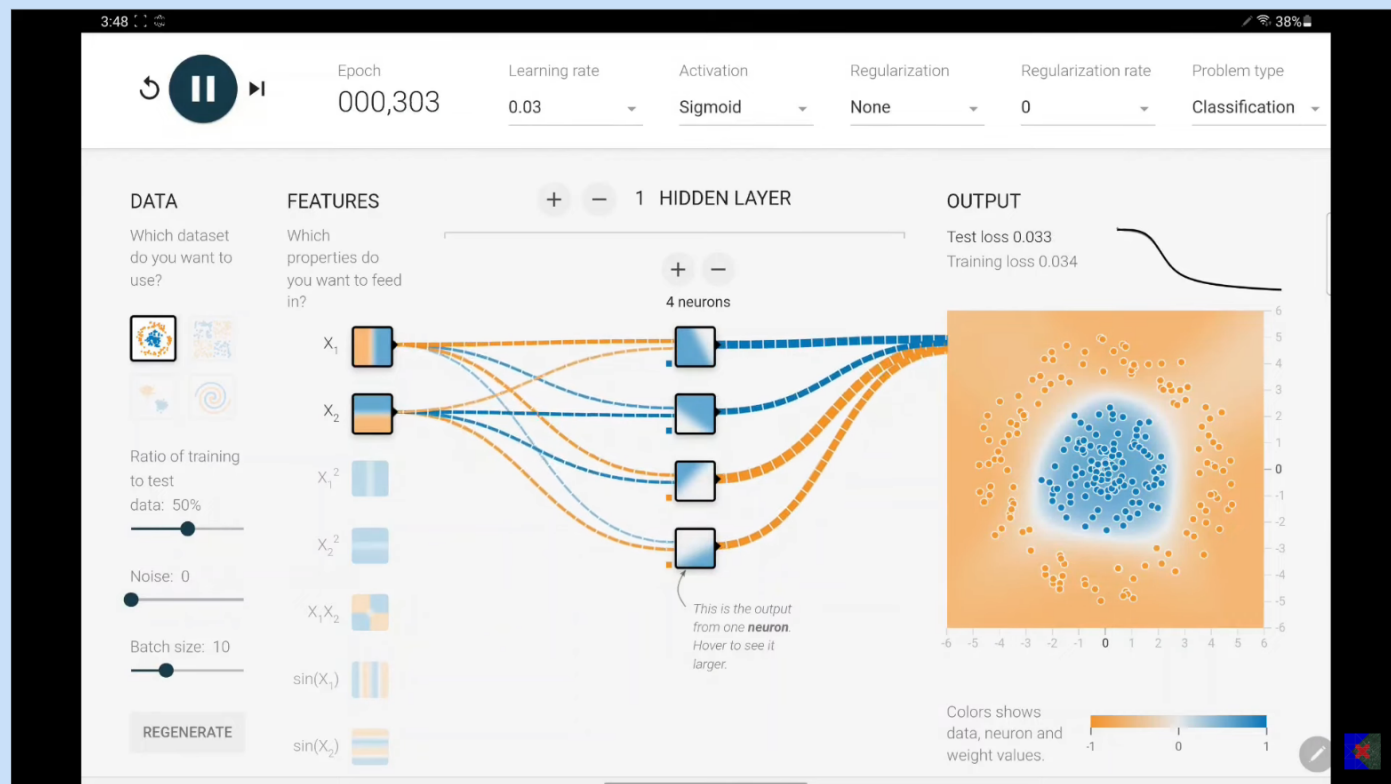
-> We can add more layers to the architecture to achieve more complex boundary line/curve

-> If the data is very complex, means the data is very non linear, we can add more hidden layer to draw a perfect boundary line.



-> Here is the same exaple of xor data that we see earlier.

->Here we just add a hidden layer with multiple nodes and we get our model that we can't get with only perceptron trick with no layer.



-> Another example.

